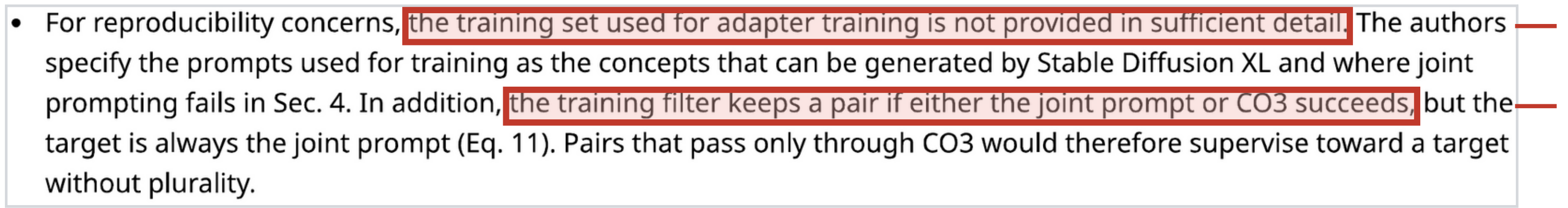
G6: The training set and the CO3 filter

PrzE weakness 2, UHz6 weakness 2 and question 1, k4nd question 3 · status: answered

THE REVIEWER'S POINT

UHz6, weakness 2



- For reproducibility concerns, the training set used for adapter training is not provided in sufficient detail. The authors specify the prompts used for training as the concepts that can be generated by Stable Diffusion XL and where joint prompting fails in Sec. 4. In addition, the training filter keeps a pair if either the joint prompt or CO3 succeeds but the target is always the joint prompt (Eq. 11). Pairs that pass only through CO3 would therefore supervise toward a target without plurality.

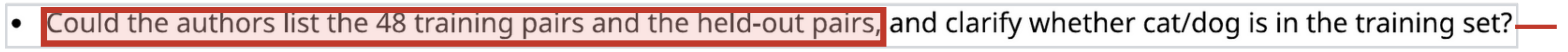
U-W2a

U-W2b

Reviews.pdf, page 3 (no text layer; passage located by OCR)

this filter

UHz6, question 1

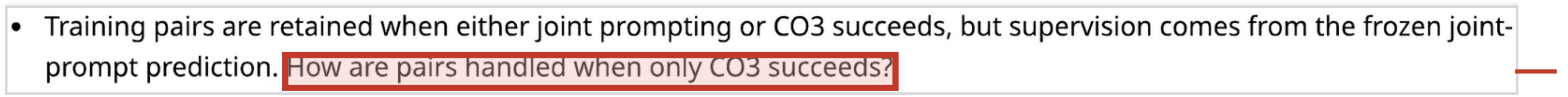


- Could the authors list the 48 training pairs and the held-out pairs, and clarify whether cat/dog is in the training set?

U-Q1

Reviews.pdf, page 3 (no text layer; passage located by OCR)

k4nd, question 3



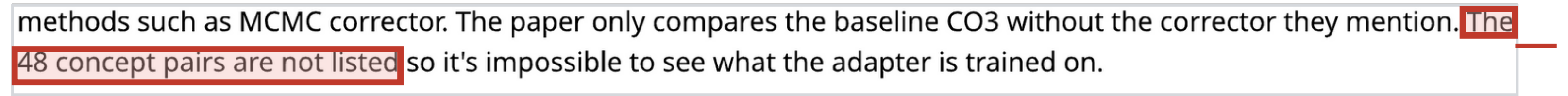
- Training pairs are retained when either joint prompting or CO3 succeeds, but supervision comes from the frozen joint-prompt prediction. How are pairs handled when only CO3 succeeds?

K-Q3

Reviews.pdf, page 5 (no text layer; passage located by OCR)

not listed

PrzE, weakness 2



methods such as MCMC corrector. The paper only compares the baseline CO3 without the corrector they mention. The 48 concept pairs are not listed so it's impossible to see what the adapter is trained on.

P-W2b

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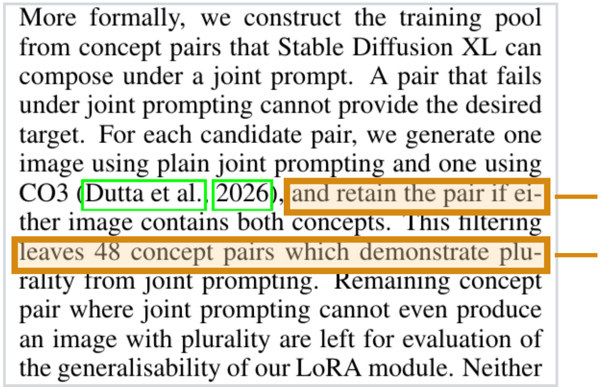
THE SUBMITTED WEAKNESS

Section 4, the training pool



More formally, we construct the training pool from concept pairs that Stable Diffusion XL can compose under a joint prompt. A pair that fails under joint prompting cannot provide the desired target. For each candidate pair, we generate one image using plain joint prompting and one using CO3 (Dutta et al., 2026), and retain the pair if either image contains both concepts. This filtering leaves 48 concept pairs which demonstrate plurality from joint prompting. Remaining concept pair where joint prompting cannot even produce an image with plurality are left for evaluation of the generalisability of our LoRA module. Neither

either image passes

48 pairs, none named

56173_Restoring_plurality_in_c.pdf (submitted), page 7

described

now

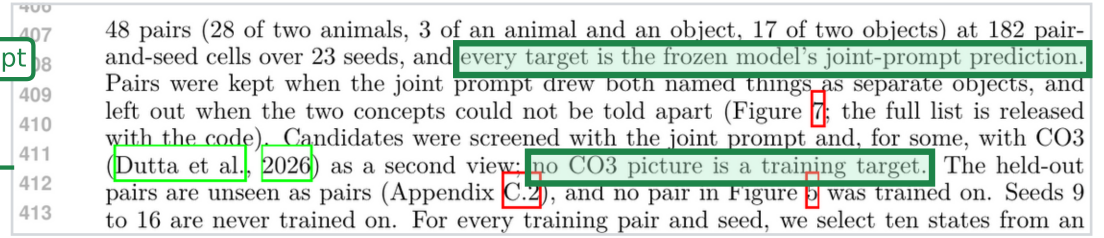
shown

THE REVISED ANSWER

Section 4, the training set

every target is the joint prompt

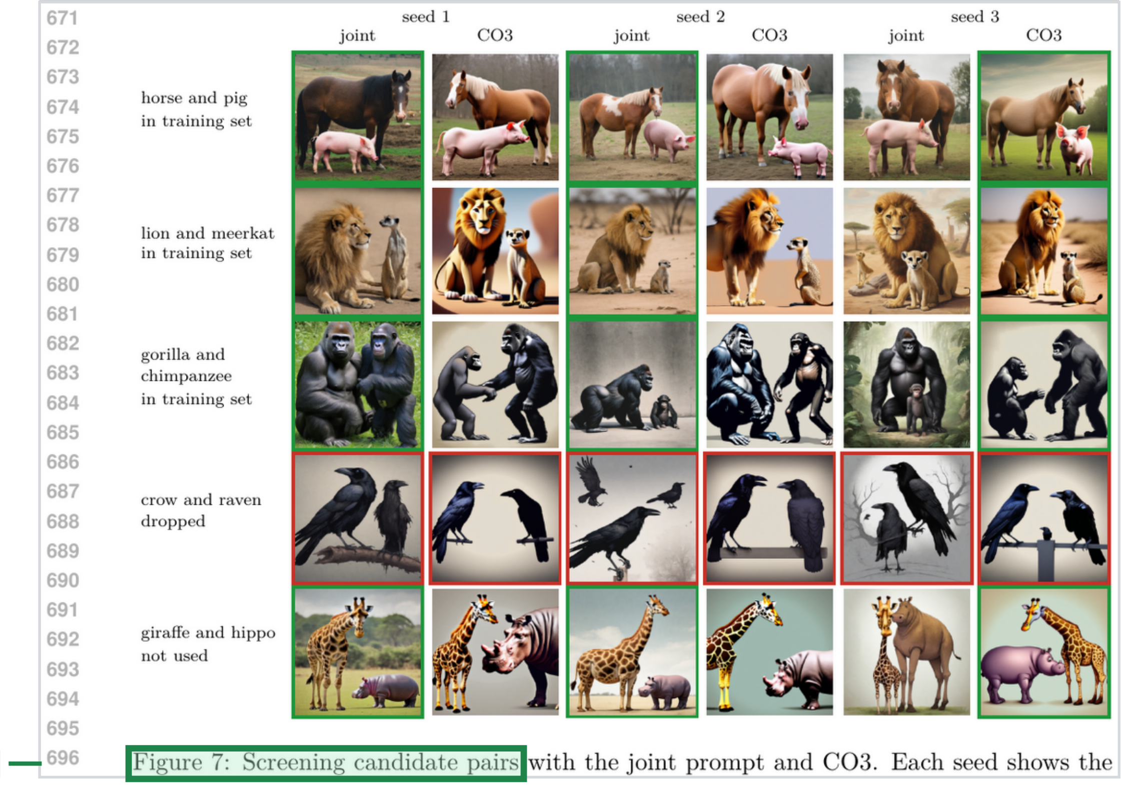
CO3 a second view only



48 pairs (28 of two animals, 3 of an animal and an object, 17 of two objects) at 182 pair-and-seed cells over 23 seeds, and every target is the frozen model's joint-prompt prediction. Pairs were kept when the joint prompt drew both named things as separate objects, and left out when the two concepts could not be told apart (Figure 7; the full list is released with the code). Candidates were screened with the joint prompt and, for some, with CO3 (Dutta et al., 2026) as a second view: no CO3 picture is a training target. The held-out pairs are unseen as pairs (Appendix C.2), and no pair in Figure 5 was trained on. Seeds 9 to 16 are never trained on. For every training pair and seed, we select ten states from an

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 8

Figure 7, screening with CO3 beside


seed 1
seed 2
seed 3
joint
CO3
joint
CO3
joint
CO3
horse and pig in training set
lion and meerkat in training set
gorilla and chimpanzee in training set
crow and raven dropped
giraffe and hippo not used

joint prompt picks the pair

Figure 7: Screening candidate pairs with the joint prompt and CO3. Each seed shows the

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 13

The reply we plan to send (replies/k4nd-and-uhz6.md: UHz6 training filter and Q1, k4nd Q3)

answered

The training filter

We have addressed this in Section 4 and Figure 7. Every training target is the frozen model's joint-prompt prediction; no CO3 picture is a target. CO3 was a second view during screening: a pair entered the training set when the joint prompt drew both named things as separate objects. Figure 7 shows the joint prompt beside CO3 for five candidate pairs, including crow and raven, dropped because the two birds are hard to tell apart. Where CO3 drew a kept pair more clearly on a seed (seed 3 in Figure 7), that seed's target is still the joint prompt's picture.

Q1: the 48 training pairs, the held-out pairs, and whether cat and dog is trained on

The training set is defined by a criterion, not by the particular pairs. A pair is kept when the frozen model's joint prompt draws both named concepts as separate objects, and dropped when the two concepts cannot be told apart (Section 4, Figure 7). The same criterion applies whether it yields 48 pairs or a thousand, and 48 is where we stopped collecting. The method relies on the criterion, not on any specific pair.

Cat and dog is not in the training set. It is held out, and no training pair contains a cat or a dog. The held-out pairs and their groups are released with the code (Appendix C.2 describes the groups), and no pair in Figure 5 was trained on.

For reproducibility, the 48 training pairs are: (two animals, 28) badger and magpie, bear and boar, bear and heron, beaver and raccoon, beaver and woodpecker, bison and weasel, buffalo and hippo, bull and rooster, camel and pelican, chicken and rabbit, deer and heron, deer and wolf, donkey and duck, donkey and rooster, emu and koala, giraffe and hyena, giraffe and ostrich, giraffe and zebra, gorilla and chimpanzee, hedgehog and robin, horse and pig, horse and sheep, impala and jackal, langur and peacock, lion and meerkat, lion and oryx, moose and chipmunk, peacock and pig; (an animal and an object, 3) hamster and carrot, pony and red barn, snail and lantern; (two objects, 17) accordion and scarecrow, bicycle and birdhouse, birdcage and hay bale, cactus and cooler box, coat rack and potted plant, guitar and traffic cone, hay bale and ukulele, lighthouse and sailboat, mailbox and pumpkin, microscope and test tube, rocking chair and cactus, soccer ball and park bench, suitcase and globe, teddy bear and birthday cake, telescope and beach ball, typewriter and cactus, windmill and tractor. They are also released with the code as the configuration file the trainer reads.

Q3: pairs retained when only CO3 succeeds

We have addressed this in Section 4 and Figure 7. Every training target is the frozen model's joint-prompt prediction; no CO3 picture is a target. CO3 was a second view during screening: a pair entered the training set when the joint prompt drew both named things as separate objects. Figure 7 shows the joint prompt beside CO3 for five candidate pairs, including crow and raven, dropped because the two birds are hard to tell apart. Where CO3 drew a kept pair more clearly on a seed (seed 3 in Figure 7), that seed's target is still the joint prompt's picture.

The 48 pairs are listed in the UHz6 Q1 reply and released with the code; the paper describes them by kind (28 two animals, 3 animal and object, 17 two objects).
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